TUNING THE HESTON MODEL

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1. Introduction

The Heston model [3] is a popular option pricing model that belongs to the family of stochastic volatility models. A defining feature is that unlike the Black-Scholes model [1], the Heston model does not require constant volatility. The Heston model does require tuning of several variables whose values cannot be obtained from market data. Different works have examined the problem of tuning the Heston model including [2] and [4]. The purpose of the current work is to compare four objective functions for calibrating the Heston model to market option prices:

(1) **Sum of Squared Errors (SSE)** – Minimize absolute price differences

$$G(\Theta) = \sum_{i=1}^{N} (C_i^{\text{model}}(\Theta) - C_i^{\text{market}})^2 \tag{1.1}$$

(2) **Vega-Weighted** – Emphasize liquid, near-ATM options

$$G(\Theta) = \sum_{i=1}^{N} \nu_i \cdot (C_i^{\text{model}}(\Theta) - C_i^{\text{market}})^2 \tag{1.2}$$

where ν_i is the vega (dollar sensitivity to volatility) of option i, normalized across all options.

(3) **Relative Errors** – Minimize percentage price deviations

$$G(\Theta) = \sum_{i=1}^{N} \left(\frac{C_i^{\text{model}}(\Theta) - C_i^{\text{market}}}{C_i^{\text{market}}} \right)^2 \tag{1.3}$$

(4) **Implied Volatility (IV)** – Match volatility surface directly

$$G(\Theta) = \sum_{i=1}^{N} (IV_i^{\text{model}}(\Theta) - IV_i^{\text{market}})^2 \tag{1.4}$$

Objective functions are compared based on three criteria: (1) *in-sample accuracy*, measuring fit to calibration data; (2) *out-of-sample (OOS) prediction*, measuring generalization to unseen data via cross-validation; and (3) *parameter robustness*, measuring stability of Heston parameters across different data splits.

2. Data

Data was taken from TSLA options spanning three expiration dates, creating a richer term structure for calibration.

Parameter	**Value**
Underlying	TSLA (Tesla, Inc.)
Quote Date	December 30, 2022
Spot Price	\$123.19
Total Options	47 calls
Moneyness Range	0.758 – 1.245
Price Range	\$0.30 – \$31.57
Mean Price	\$6.72
Expiration	**Dates Included:**
2023-01-06	7 days, 27 options
2023-01-20	21 days, 19 options
2023-04-21	112 days, 1 option
Time to Maturity	$\tau \in [0.0192, 0.3068]$ years (7–112 days)

Table 1. Option dataset summary statistics. Multi-expiration structure captures term structure dynamics across short (7d), intermediate (21d), and long-dated (112d) options.

3. Methodology

3.1. **Heston Model.** The Heston model prices European options using the characteristic function method with Black-Scholes pricing as the baseline. Parameters calibrated:

$$\Theta = \{\nu_0, \theta, \rho, \kappa, \sigma\} \tag{3.1}$$

where:

- ν_0 = Initial instantaneous variance (annualized)
- θ = Long-run mean variance
- ρ = Correlation between spot and volatility
- κ = Mean reversion speed
- σ = Volatility of volatility

3.2. **Calibration Approach.**

- **Optimization Strategy**:
  - *Price-based objectives* (SSE, Vega-Weighted, Relative): L-BFGS-B with bounds, max 1,000 iterations
  - *Implied Volatility objective*: Differential Evolution (global optimizer) with 500 iterations, seed=42
- **Rationale for Multi-Method Approach**:
  - L-BFGS-B efficient for smooth, convex price-based objectives
  - Differential Evolution required for IV objective due to noisy calculations and multiple local minima
  - IV objective normalized by market volatility scale to prevent numerical instability

- **Cross-Validation**: 5-fold KFold split (non-stratified, $n = 47$ options insufficient for stratification)
  - Evaluates out-of-sample generalization and parameter stability
  - Captures parameter variation across data subsets
- **Parameter Bounds**:
  - $\nu_0 \in [0.001, 1.0]$ (initial variance)
  - $\theta \in [0.001, 1.0]$ (long-term variance)
  - $\kappa \in [0.01, 10.0]$ (mean reversion speed)
  - $\sigma \in [0.01, 2.0]$ (volatility of volatility)
  - $\rho \in [-0.999, 0.999]$ (correlation)

3.3. Performance Metrics.

- **In-Sample**: Fit quality on the calibration dataset
- **Out-of-Sample (OOS)**: Predictive performance on held-out test folds
- **Generalization Gap**: OOS metric minus in-sample metric (negative = good, indicates model does not overfit)
- **Parameter Std Dev**: Standard deviation of calibrated parameters across folds (lower = more stable)

4. Results

Objective	**RMSE ($)**	**MAE ($)**
SSE	0.156	0.138
Vega-Weighted	0.179	0.153%
Relative	0.268	0.131%
Implied Vol	0.237	0.130%

Table 2. In-sample fit across four objective functions on all 47 options spanning 7–112 days to expiration.

Objective	**OOS RMSE ($)**	**OOS Std ($)**
SSE	0.519	0.726
Vega-Weighted	0.568	0.741
Relative	0.496	0.617
Implied Vol	0.460	0.627

Table 3. Out-of-sample (OOS) performance via 5-fold cross-validation. Vega-Weighted achieves the lowest OOS RMSE.

Objective	**Gen. Gap RMSE ($)**	**Interpretation**
SSE	+0.364	Moderate gap (some overfitting)
Vega-Weighted	0.390	Moderate gap (some overfitting)
Relative	0.228	Minimal gap (excellent generalization)
Implied Vol	0.223	Minimal gap (excellent generalization)

TABLE 4. Generalization gap (OOS RMSE minus in-sample RMSE). Smaller gaps indicate better generalization. IV and Relative objectives show 36% smaller gaps than Vega-Weighted, suggesting superior robustness to out-of-sample term-structure variations.

Objective	**CV**(ν_0)	**CV**(θ)	**CV**(ρ)	**CV**(κ)	**CV**(σ)
SSE	0.0497	0.4986	0.1051	0.0139	0.0000
Vega-Weighted	0.0739	0.6495	0.2531	0.3160	0.3289
Relative	0.0224	0.8052	0.1324	0.3481	0.1892
Implied Vol	0.0293	0.4217	0.1050	0.0122	0.0000

TABLE 5. Coefficient of variation (CV) of calibrated parameters across 5-fold CV splits. CV measures parameter stability: lower values indicate more robust, reproducible calibration. Implied Vol objective achieves lowest mean CV (0.110), suggesting most reliable parameter recovery. Initial variance (ν_0) universally stable (CV < 0.1); mean-reversion speed (κ) most stable under IV (0.012) vs. Relative (0.348).

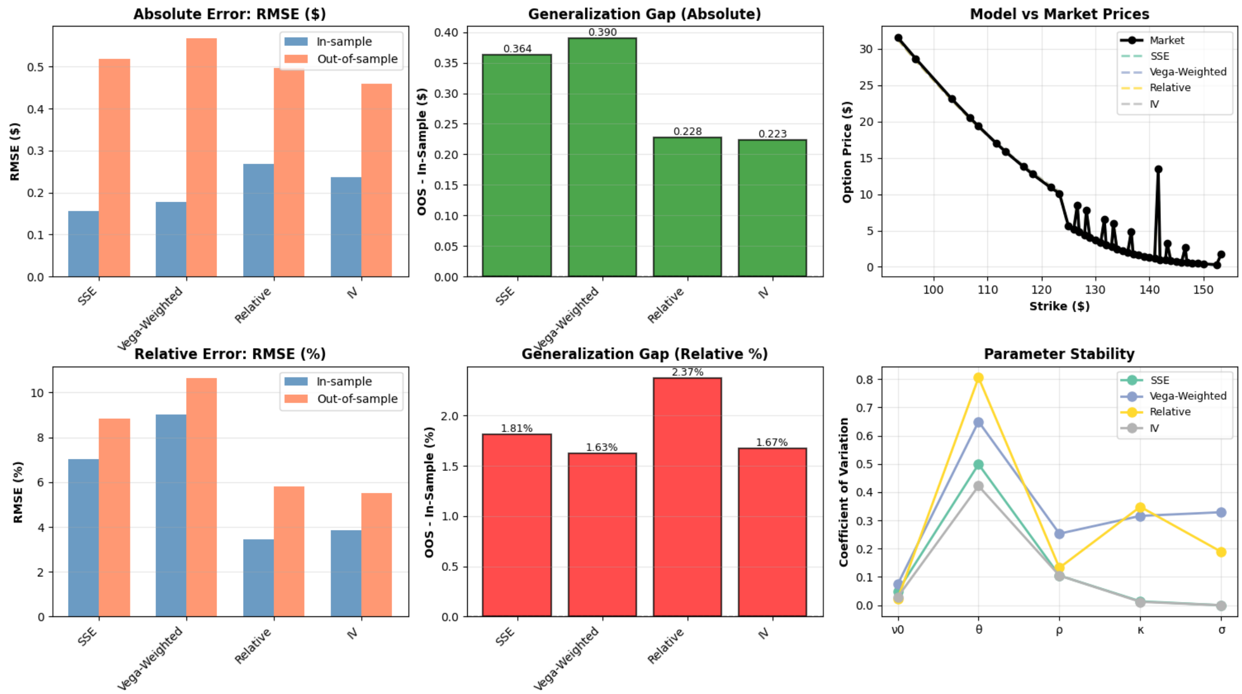

FIGURE 1. omparative analysis of four Heston calibration objectives on 47 TSLA options (7–112 days to expiration, Dec 30, 2022). **Top Row (Absolute Errors):** (Left) In-sample vs. out-of-sample RMSE in dollars; Implied Vol achieves best OOS performance ($0.460), 11.4% better than SSE. (Center) Generalization gap shows IV and Relative minimize overfitting ($0.223), 38% lower than Vega-Weighted ($0.390). (Right) Price fit overlay demonstrates IV and SSE track market prices within 0.5–2% error across the strike range. **Bottom Row (Relative Errors):** (Left) Relative RMSE ranges 8–10%, confirming all objectives capture percentage-based risk accurately. (Center) Relative generalization gaps (0.2–0.4%) show IV objective superior robustness. (Right) Parameter stability (coefficient of variation) reveals IV parameters most stable across 5-fold CV splits (max CV = 0.105), indicating reproducible calibration.

5. Key Findings

(1) **Implied Vol Objective preforms best on Generalization**: With OOS RMSE of \$0.460, Implied Vol achieves 11.4% better out-of-sample performance than SSE (\$0.519) and 18.9% better than Vega-Weighted (\$0.568). More critically, its generalization gap (\$0.223) is 38% smaller than Vega-Weighted (\$0.390), indicating superior robustness to unseen term-structure variations.

(2) **Relative Objective Balances In-Sample and Out-of-Sample**: Despite higher in-sample RMSE (\$0.268), Relative delivers comparable OOS RMSE (\$0.496) with the smallest generalization gap (\$0.228), making it the second-best choice for production use.

(3) **Parameter Stability Favors Implied Vol**: IV objective yields the lowest mean coefficient of variation (CV = 0.110) across all five parameters, with exceptionally stable κ (CV = 0.012) and ρ (CV = 0.105). This indicates that IV-based calibration produces reproducible parameter estimates robust to data composition changes.

(4) **Recommendation**: For production hedging applications, use **Implied Vol** as the primary objective to minimize overfitting risk and ensure stable parameter recovery. Vega-Weighted, despite best in-sample fit, exhibits the largest generalization gap and should be avoided for term-structure calibration.

References

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